Blockchain Practice PS

There are four possible tracks in this task as described in the coming sections. You can choose and work on any of those

1. Aptos MetaMask Snap (Simplified)

- **Task**: Build a minimal MetaMask Snap prototype that displays basic account info (e.g., balance) from both an EVM-compatible chain and the Aptos blockchain.
- **Requirements**:
  - Your understanding of **Aptos MetaMask Snap** and its use cases.
  - Implement a basic Snap that connects MetaMask to Aptos and retrieves the user's wallet balance on both blockchains.
  - Provide brief documentation on the setup and usage.
- **Deliverables**: Code repository with instructions for running the Snap and a brief demo video.

2. Zero Day Futures Contracts (MVP)

- **Task**: Design a simple decentralized contract on Aptos that simulates short-term futures trading.
- **Requirements**:
  - Your understanding of **Zero Day Futures Contracts** and its use cases.
  - Create a smart contract for a 24-hour futures contract that locks in an initial balance and settles based on a basic condition (e.g., simulated price increase/decrease).
  - Include a frontend that shows basic interaction with the contract (e.g., placing a mock futures trade, viewing settlement results).
- **Deliverables**: Smart contract code, a frontend demo page, and a README explaining setup and usage.

3. Wordle/Qwordle On-Chain (Mini Version)

- **Task**: Develop a simple on-chain version of Wordle with limited functionality.
- **Requirements**:
  - Your understanding of **Wordle/Qwordle On-Chain** and its use cases.
  - Create a contract that accepts a user’s 5-letter word guess and validates it against a pre-set solution.
  - Provide a basic frontend to submit guesses and view results.
- **Deliverables**: Smart contract code, a basic frontend interface, and documentation.

4. On-Chain Radio (Prototype)

- **Task**: Design a basic on-chain radio feature to upload and play short audio snippets.
- **Requirements**:
  - Your understanding of on-Chain Radio and its use cases.
  - Allow users to upload a short audio file (or link) to a blockchain-based database.
  - Implement a payment simulation where users can tip for each play.
  - Provide a simple frontend where users can upload, listen, and tip.
- **Deliverables**: Codebase for the contract and frontend, with a README and setup instructions.

SUBMISSION & EVALUATION CRITERIA

Submission Deadline: 29th October, 2026

Evaluation Criteria:

- You are expected to pick **one** of the above 4 listed possible tracks, and work on it.

If you pick more than a single track from above, the **best one** out of all the tracks chosen by you will only be considered for evaluation.